SADPmini2-Ex Hand-Held Dew Point Meter

Next Generation Intrinsically Safe Hand Held Dewpoint Meter

Rugged, light weight IP66 construction makes **Model SADPmini2-Ex** the most reliable dewpoint hygrometer available for rapid spot checks of dewpoint or trace moisture content in most gases and compressed air, where a certified instrument is required. The ultra-high capacitance sensor gives unsurpassed sensitivity, speed of response, repeatability and stability.

Model SADPmini2-Ex is ideal for use in hazardous areas.

- ATEX IEC IECEx c UL US LISTED



USB	☑
BLUETOOTH	☑
LOGGING	☑
SD CARD	☑
LI-ION BATTERY	☑
CALCULATOR APP	☑
2 YEAR WARRANTY	☑

Fast Repeatable Measurements

Reduced volume of the redesigned stainless steel sample chamber gives rapid spot check measurements of moisture in gases and dry compressed air.

The unique innovative keypad design allows quick, intuitive, selection of the versatile features incorporated in the instrument and modification of all set up choices. The full colour LCD display shows the measurement in two, independently selectable, units simultaneously, together with a linear analogue representation of the main unit. A single button press changes the display to a real time graph of the readings in the main units.

Connection to a PC/Laptop via USB or Bluetooth also enables the user to modify and save the set up and save details via the PC. Powerful logging and graphing capabilities and Bluetooth communication are standard features. A mobile app is available for display of readings on smartphone or tablet.

KEY FEATURES

- ATEX IEC IECEx c UL US LISTED Certified Intrinsically safe for use in hazardous areas to ATEX, IECEx & cULus Listed.
- Full colour graphical display showing multiple units simultaneously.
- Desiccant Dry-Down Chamber for faster response by keeping sensor dry between tests.
- AutoCal Span Correction, for optimum accuracy between laboratory calibrations.
- Measurement units selectable in °C or °F dewpoint, ppm(v), ppm(w), g/m^3 or lb/MMSCF.
- User selectable display in choice of 10 languages: English, French, German, Spanish, Italian, Portuguese, Russian, Chinese, Japanese & Korean.
- Flexible programming for Data Logging with capacity in excess of 300,000 values.
- Real time logging/graphing of results with Quick Log and Snapshot features.
- Wireless Bluetooth & USB connectivity for configuration & data upload/download to pc or laptop.
- View display screen on smart phone, or tablet, via mobile App.
- Wirelessly print results to dedicated Bluetooth printer.
- Integral pressure calculator to display pressure dewpoints.
- Easy to use, with icons, intuitive control and latest user interface.
- Compact ergonomic design with a strong, durable and rugged body.
- Rechargeable batteries, allowing in excess of 150 hours use in continuous operation.
- SD micro-card installed for data and settings backup, user manual retrieval and calibration history.
- User ability to update latest firmware from factory.
- Includes Certificate of Test & Calibration, referenced to National & International Standards NPL/NIST.
- Anti-static carry bag included.

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Desiccant Dry Down Technology

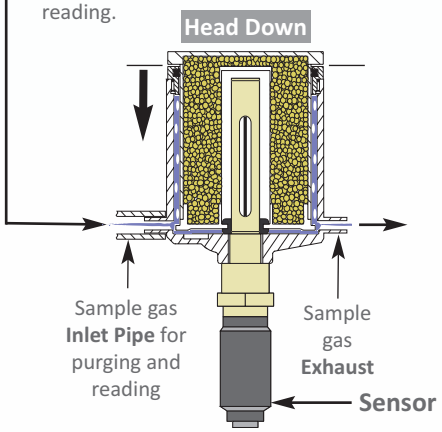
The Desiccant Head Assembly

Keeping the sensor dry between tests ensures that the **SADPmini2-Ex** is always ready to carry out rapid spot checks. The unique design of the Desiccant Head achieves this by surrounding the sensor with desiccant before the head is raised for sampling.

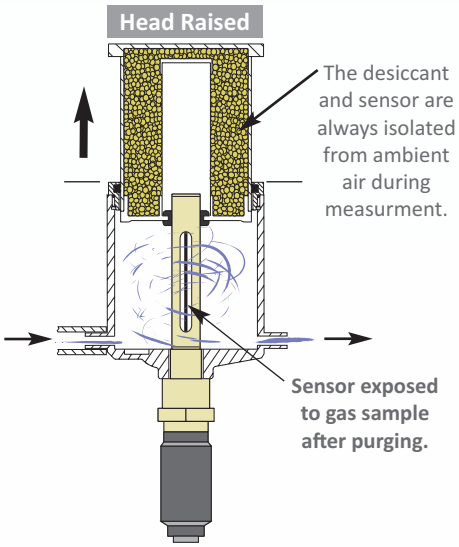
At no time is the sensor allowed to come into contact with ambient air. The chamber is also designed so that the void space and chamber wall surfaces are purged with sample gas, before exposure of the sensor, so giving faster, more accurate and reliable results.

The unique design allows purging of the gas in the "**Dry-Down Position**" which will bring all surfaces and voids to equilibrium.

The "**Head**" can then be raised to take a reading.



Dry-down position



After display has settled, take a reading

Specifications

Hazardous Area Certification

II 1G Ex ia IIC T4 Ga (-20°C ≤ Ta ≤ +50°C)
II 1D Ex ia IIIB T60°C T_{50}80°C Da (-20°C ≤ Ta ≤ +50°C)

Ex ia IIC T4 Ga (-20°C ≤ Ta ≤ +50°C)
Ex ia IIIB T60°C T_{50}80°C Da (-20°C ≤ Ta ≤ +50°C)

INTRINSICALLY SAFE / SÉCURITÉ INTRINSÈQUE Exia
Class I Div 1 Groups A, B, C and D T4
-20°C ≤ Ta ≤ +50°C

Sensor type
Ultra-high capacitance aluminium oxide.
Ranges
-110°C to -20°C (-166°F to -4°F) dewpoint, Silver
-100°C to 0°C (-148°F to +32°F) dewpoint, Purple
-80°C to +20°C (-112°F to +68°F) dewpoint, Blue
Calibration
Supplied with a Certificate of Test and Calibration traceable to NPL/NIST.
Accuracy
Better than ±2°C dewpoint (±3.6°F)
Repeatability
Better than ±0.3°C dewpoint (±0.54°F)
AutoCal
Span check and correction carried out by following simple on screen instructions.
Temperature coefficient
Temperature compensated for operating range.
Typical sensor response times
Dry to Wet: -110°C to -20°C dewpoint, <20 secs
Wet to Dry: -10°C to -60°C dewpoint <180 secs
Sample flow
Flow independent, recommended 5 to 10 lt/min, max 20 lt/min.
Sensor life
Typically 7-10 years, dependant on application.
Pressure dewpoints
Integral calculator for display of dewpoints at pressure for both "ideal" gases and natural gas.

Electrical

Electromagnetic compatibility
Conforms to EMC Directive 89/336/EEC, amended 95/31/EEC
Display
3.5 inch full colour LCD graphical display with backlight.
Power supply
Rechargeable Li-ion battery.
Charges from USB or universal mains charger supplied.
Battery life
In excess of 150 hours continuous use on full charge.
Time to full charge
Empty to full, approx. 14 hours.
Warm up time
2 seconds
Micro SD card
Used as a data log backup. Also contains User Tool software and pdf of Operating Manual.

Mechanical

Weight
1.4 kg (3.1 lb)
Dimensions
215 x 108 x 124 mm (8.47 x 4.25 x 4.88 inches)
Ingress protection
IP66/NEMA4X
Operating pressure
Atmospheric pressure. 0.3 barg (4 psig)
Operating temperature (ambient)
-20°C to +50°C (-4°F to +122°F)
Operating humidity (ambient)
Max. 95% non-condensing
Storage temperature and humidity
-20°C to +50°C (-4°F to +122°F) Max. 95% non-condensing

Warranty

24 months faulty workmanship and materials.

Logging

Integral data storage
Up to 300,000 date and time stamped readings
Stored in up to 20 user definable folders.
Logging frequency
User definable from 5 seconds to 1 hour.
Data display
Graphical and numerical
Real time logging
Can log directly to PC when connected via USB or Bluetooth.
Bluetooth
Bluetooth connectivity provided with logging package.

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SADPmini2-Ex

Dimensions



How to Order

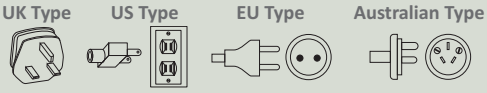
Model SADPmini2-Ex - [XX] - [F] - [X] - [XX]

RANGE IN DEW POINT:
SR: -110°C to -20°C (-166°F to -4°F)
PL: -100°C to 0°C (-148°F to +32°F)
BL: -80°C to +20°C (-112°F to +68°F)

F: Stardard Supply - Fir Tree Fitting

4: 0.25" (1/4") - Swagelok SS Compression Type
6: 6 mm - Swagelok SS Compression Type
8: 0.125" (1/8") -Swagelok® SS Compression Type

MAINS/AC PLUG TYPE - For Charger = XX
UK: Standard UK, 3 Square pin
US: 2 Vertical Flat Blade
EU: 2 Round Pins
AU: 2 Oblique Flat Blades with Ground



Fitting 1: Fir Tree Fitting
Fitting 2: Compression Fitting



EXAMPLE:
To order this instrument with a range of -80°C to +20°C dewpoint, 6mm compression fitting, and a UK AC power plug, the order code is:- **Model SADPmini2-Ex - BL - F - 6 - UK**

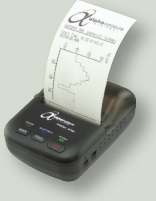
Standard Accessories (Included with each instrument)

Included:
Anti-static carry bag with adjustable shoulder strap, 2m PTFE sample tube, adjustable lanyard, micro USB to USB communication battery charger cable, universal 110/240 Vac charging plug, 4GB micro SD card, illustrated operating manual.



Optional Extras

Bluetooth Printer.



Portable Sample System.



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Notes

Authorised Distributor Information

Note: Model SADPmini2-Ex is ideal for most industrial gases, but is not suitable for use in the presence of Ammonia (NH_3), Chlorine (Cl_2), Hydrogen Chloride (HCl), Hydrogen Fluoride (HF), Nitrous Oxide (NO_2) and its hydrolysed form, Nitric Acid (HNO_3). Sulphur Dioxide (SO_2) can be monitored if the moisture content is low, less than 100ppm(v).

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Product specification may be subject to change.

CERTIFICATE No. FM35600
BS EN ISO 9001:2008

Registered in England and Wales No. 3902302
VAT Registration No. GB 607 2075 63
WEEE Producer Registration No. WEEE/EA0067TX

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